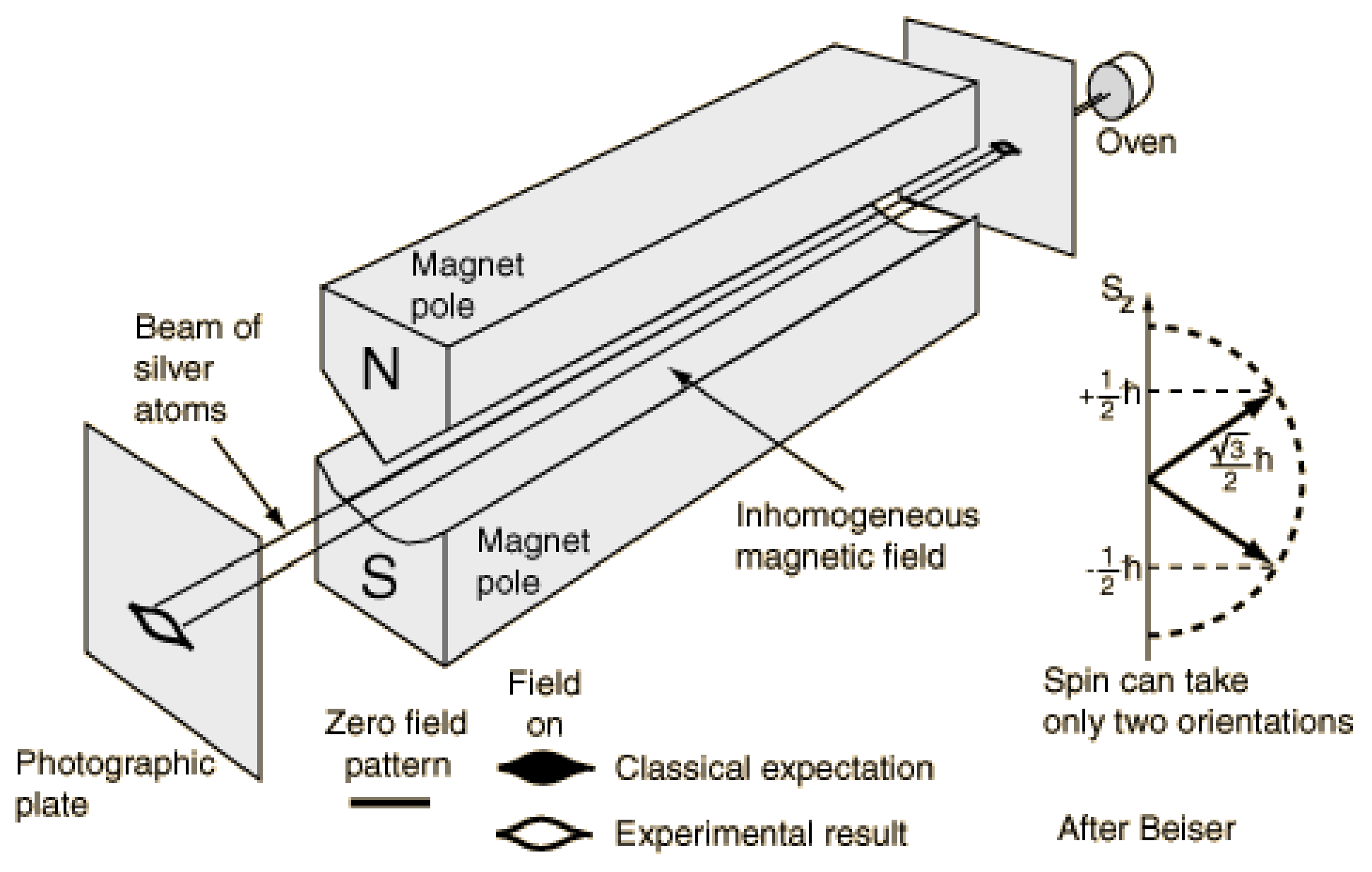

Oven
Magnet pole
N
Beam of silver atoms
S
Magnet pole
Inhomogeneous magnetic field
Photographic plate
Zero field pattern
Field on
Classical expectation
Experimental result
S_z
$+\frac{1}{2}\hbar$
$\frac{\sqrt{3}}{2}\hbar$
$-\frac{1}{2}\hbar$
Spin can take only two orientations
After Beiser